Machine Learning Project Report: Titanic Survival Prediction

1. Project Objective

The primary objective of this project is to build a predictive machine learning model that determines whether a passenger survived or perished during the Titanic shipwreck of 1912.

By analyzing individual passenger attributes—such as age, gender, socio-economic status (passenger class), ticket fare, and family relationships—the project aims to uncover the underlying patterns that dictated survival. Ultimately, this classification problem serves to evaluate and compare the predictive performance of linear models (Logistic Regression) against non-linear, rule-based structures (Decision Trees).

2. The Machine Learning Process Workflow

The development lifecycle of this predictive model follows a structured, four-phase machine learning pipeline.

Phase 1: Data Preprocessing & Cleaning

Raw datasets are rarely ready for mathematical modeling. This initial phase focuses on handling missing data and converting variables into formats readable by machine learning algorithms.

- **Imputation of Missing Variables:** * **Age:** Missing records are replaced with the *median* age of the dataset. The median is chosen over the mean because it is robust against outliers (extreme age gaps), preventing skewness.
    - **Embarked:** Missing ports of embarkation are filled with the *mode* (the most frequently occurring port).
    - **Cabin:** Because a massive percentage of cabin numbers are missing, the vacant fields are filled with the placeholder label "Unknown". This prevents the loss of data rows and treats "missingness" as an independent characteristic.
- **Feature Engineering:**
    - **Family Size:** A new metric is engineered by combining SibSp (siblings/spouses) and Parch (parents/children). This condenses family tracking into a single variable to evaluate if traveling solo or in a group altered survival odds.

    - **Title Extraction:** Text names are unique and uninformative for algorithms. However, extracting structural titles (*Mr., Mrs., Miss., Master*) via text patterns creates a strong feature capturing both social standing and age brackets.
- **Categorical Encoding:** Machine learning algorithms process numerical equations, meaning text descriptions like "male" or "Southampton" must be translated. **Label Encoding** converts these textual categories into distinct integers (e.g., assigning 0 or 1), making them mathematically viable for the models.

Phase 2: Exploratory Data Analysis (EDA)

EDA is the diagnostic phase used to identify patterns, anomalies, and correlations between passenger attributes and their final survival status before any models are built.

- **Demographic Analysis:** Mathematical grouping establishes a massive statistical disparity between genders. The survival rate for female passengers is exponentially higher than for male passengers, validating the historical "women and children first" maritime protocol.
- **Socio-Economic Stratification:** Aggregating survival rates across passenger classes (Pclass) reveals a stark divide. First-class passengers enjoyed a vastly superior survival probability compared to third-class passengers, highlighting class privilege during the evacuation.
- **Visual Data Profiling:** * **Baseline Ratios:** Count plots map out the baseline class balance, revealing that a larger portion of passengers perished than survived.
    - **Distribution Analysis:** Histograms map the age distribution of the ship, highlighting that the bulk of the passengers were young to middle-aged adults, while also identifying clusters of children who were prioritized during rescue efforts.

Phase 3: Model Planning

Model planning bridges data exploration and algorithmic training. It ensures the data is partitioned properly to prevent biased evaluations.

- **Feature Isolation:** Irrelevant or purely administrative features (such as PassengerId, structural Names, and alphanumeric Ticket tokens) are dropped. The independent features () are isolated from the target variable (, Survived).
- **Data Partitioning (Train-Test Split):** The dataset is split into an **80/20 ratio**.

    - **Training Set (80%):** Fed into the algorithms so they can discover relationships between passenger features and survival.
    - **Testing Set (20%):** Kept completely unseen by the models. It serves as an unbiased baseline to test how well the trained models generalize to completely new data.
- **Algorithmic Strategy:** Two distinct classification strategies are selected:
    1. **Logistic Regression:** A linear classifier that uses the sigmoid function to calculate the probability of an binary outcome. It establishes a baseline performance.
    2. **Decision Tree Classifier:** A non-linear, hierarchical model that splits data into sequential branch rules. It excels at capturing combined feature relationships (e.g., *"If female AND in 3rd class..."*).

Phase 4: Model Building & Evaluation

The final phase involves exposing the algorithms to the training data, tuning their limits, and measuring performance using exact error metrics.

- **Model Optimization:** * For **Logistic Regression**, optimization limits are expanded (max_iter=1000) to guarantee that the underlying solver converges on the most stable mathematical boundary.
    - For the **Decision Tree**, a regularizing cap (max_depth=4) is strictly applied. Restricting tree depth prevents the model from mapping out noise and memorizing the training set, a flaw known as **overfitting**.
- **Performance Diagnostics:** Models are evaluated on the unseen testing set using two metrics:
    - **Accuracy Score:** The raw percentage of total correct predictions.
    - **Confusion Matrix:** A critical analytical grid that segments predictions into True Negatives (correctly predicted deaths), True Positives (correctly predicted survivals), False Positives (Type I errors), and False Negatives (Type II errors).

3. Conclusion & Key Takeaways

1. **Data Quality Impacts Outcomes:** Basic data preprocessing—such as median imputation and label encoding—is vital. Without structural transformations, machine learning models cannot interpret messy historical data.

2. **Feature Engineering Drives Signal:** Synthesizing hidden indicators (like pulling social titles out of raw names) provides models with clean, highly predictive markers that vastly improve classification accuracy over raw data alone.
3. **Linear vs. Rule-Based Trade-offs:** * **Logistic Regression** acts as a dependable baseline, performing well when trends are distinct and linear (such as the overwhelming survival rate of women over men).
    - **Decision Trees** offer deeper flexibility by mapping complex, multi-layered passenger profiles. However, they require careful boundary capping (max_depth) to stay generalized.
4. **Historical Validation:** Ultimately, both machine learning workflows mathematically validate historical accounts of the disaster: socio-economic class, fare paid, and gender were the absolute defining catalysts for survival on the Titanic.